

Free-View XR Tours Using 3D Gaussian Splatting with an Avatar Guide

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Figure 1: (a) Point-cloud from Structure-from-Motion (SfM), (b) Testing Gaussian-primitives in Unity, (c) Loading and previewing 3DGS scenes in the Unity, (e,f,g) In-VR visualization of reconstructed scenes on the Meta Quest 3, (d,h) Case-Study

ABSTRACT

XR tours are often built from manual 3D authoring, 3D scanning, photogrammetry, or fixed-view 360° video, which can be time-consuming to produce or limit free exploration. 3D Gaussian Splatting (3DGS) offers photoreal free view rendering from captured imagery, yet its use in end-to-end XR touring experiences, especially with lightweight in-scene guidance remains underexplored. We present an XR tour system for free-view exploration of 3DGS scenes in XR that combines controller-based 6-DoF navigation with an avatar guide that delivers proximity triggered text-to-speech narration. We demonstrate the system in a public setting and report observations from a case study with >50 participants.

Index Terms: 3D Gaussian Splatting; free-view navigation; XR tour; 3D avatars;

1 INTRODUCTION

XR tours are increasingly popular in museums, education, tourism, and cultural heritage. However, many of them depend on either manually authored 3D environments or fixed-view representations (e.g., 360° video), which are costly to produce at scale or restrict users' ability to explore freely. Capture-based pipelines offer an alternative by reconstructing navigable 3D content directly from real-world imagery. In particular, 3D Gaussian Splatting (3DGS) supports photoreal free-view rendering, providing a promising foundation for practical, free-view XR touring experiences [4].

Prior work has established foundations for immersive viewing and tele-navigation in captured 3DGS scenes. VRSplat addresses VR-specific 3DGS rendering issues, including stereo stabil-

ity and performance constraints on head-mounted displays (HMD) [7]. EyeNavGS introduces record-and-replay tools and datasets for studying 6-DoF navigation behaviour in real-world 3DGS scenes in VR [3]. Together, these efforts strengthen the basis for immersive 3DGS viewing and tele-navigation. In parallel, Kwon and Yu present a 3DGS-based virtual museum and compare it with a 360° panorama baseline, reporting higher presence with free-view navigation [5]. Their evaluation is conducted in a desktop setting (mouse/keyboard), without an HMD-based VR study. Works such as CeylonXR uses 3DGS-based exploration with AI guidance features [8], while our work presents an avatar based information delivery. To the best of our knowledge, free-view XR tour exploration over captured 3DGS scenes that integrates in-scene information delivery through an avatar guide remains underexplored.

We present a free-view XR experience for captured 3DGS scenes in an iconic-spot touring scenario, where users can look around and move freely in the surrounding space to support telepresence. Users explore with controller-based 6 DoF navigation in an HMD. Proximity triggered text to speech narration is delivered by an in-scene avatar, which serves as a spatial reference point for contextual information and activates implicitly as users approach. This supports in-scene information delivery during free-view exploration without additional menus or explicit selection. We report a public facing walk up case study with over 50 participants and summarise practical insights for deploying guided free-view XR experiences over captured reconstructions. Figure 1 shows the outputs of our work.

2 SYSTEM DESIGN AND IMPLEMENTATION

We frame our XR tour scenario centred on an iconic spot, where users are able to look around and move freely in its surroundings. 3D Gaussian Splatting (3DGS) is well-suited to this setting, providing photoreal free-view navigation from multi-view imagery with real-time rendering. Using an HMD and controllers, our system enables users to explore around the spot with 6-DoF viewing in XR, aiming to enhance the sense of telepresence during the tour.

Figure 2 summarises the workflow from capture to VR playback. We captured six iconic spots covering indoor and outdoor scenes, using a smartphone camera with manual settings that lock exposure, focus, and white balance to prevent automatic changes

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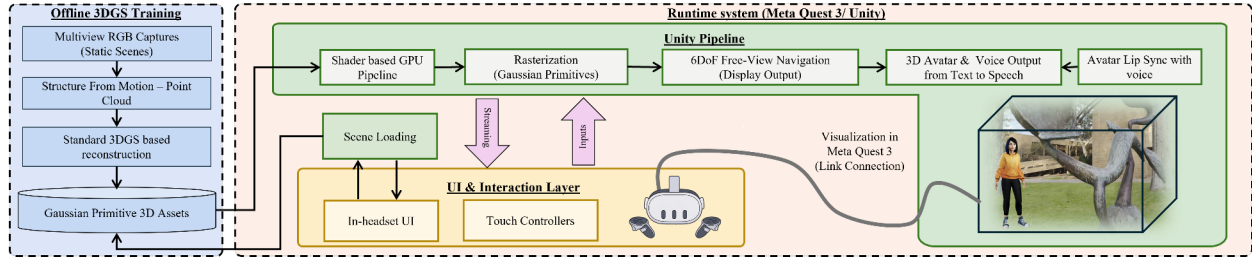


Figure 2: End-to-end workflow for captured 3DGS scene preparation and avatar-assisted free-view exploration in VR.

between shots and support SfM feature matching and camera registration. We captured around 500 images per location on average (300–800 depending on scene size and texture) to ensure sufficient multi-view coverage and overlap for estimating camera intrinsics and poses, collecting overlapping views while moving around an anchor object (e.g., a statue or tree) so shared features strengthen SfM correspondences. We also compared extracting frames from video with manual photo capture in 3DGS reconstruction, and observed higher visual fidelity with photos, since subtle motion blur in video frames limits feature matching for SfM. Each location was then reconstructed with a standard 3DGS training workflow [4].

For VR visualisation, 3D Gaussian primitives are streamed into Unity runtime and rasterised using a shader-based GPU pipeline [6]. It is rendered on a Meta Quest 3 via Link connection (Figure 2) and runs at 60 FPS during exploration for scenes averaging 1M Gaussian primitives. Users explore each scene using touch controller-based free-view navigation: the controllers drive view-point translation and yaw rotation, while head pose continuously controls view orientation for natural, self-directed viewing.

An in-headset scene-selection UI (Figure 1(e)) enables switching between reconstructions and triggers scene loading, which is useful when users explore multiple locations. For in-scene guidance, a 3D avatar guide provides proximity-aware audio narration: text-to-speech audio plays when the user enters a configurable radius, using distance-to-avatar as the interaction cue and avoiding menus, explicit selection, or constrained paths. Avatars were created using Ready Player Me and imported into Unity [2]. Oculus LipSync for Unity drives mouth and lip movement from the narration audio [1].

3 CASE STUDY

We conducted a case study during a public event with >50 participants across a wide age range (approximately 10–60 years old). Participants wore the HMD, received a short explanation of the controls, and explored the selected scenes with free-view navigation as in Figure 1(d,h). An avatar guide appeared in each scene, triggering narration when participants entered the proximity radius.

After each session, we gathered participant feedback through brief post-session interviews. Participants commonly reported a strong sense of immersion with telepresence, and several who had previously visited the captured locations noted that key visual cues and spatial layout felt recognisable, with movement through the reconstruction resembling walking through the real place. Participants described free-view navigation as giving them agency to explore, while avatar-triggered narration provided contextual information that guided where to look during exploration. They also pointed to potential uses such as museum and education tours, indoor wayfinding via a spatial preview alongside a 2D map layout, and game-like experiences within the captured environments, especially for younger users. Some participants also expressed interest in continuous movement across a wider environment range, such as navigating multiple rooms within the same place.

4 CONCLUSION

We present a free view XR tour experience for exploring captured real-world environments using 3DGS, combining controller-based 6 DoF navigation with an in scene avatar guide that delivers proximity triggered text to speech narration. Feedback from the case study suggests a strong sense of immersion with telepresence, with several participants recognising familiar visual cues and spatial layout, and that avatar-triggered narration guided attention while providing place specific context without additional menu interaction.

Our prototype has several limitations. Visual artefacts may appear in 3DGS reconstructions, often attributable to limited capture samples (e.g., blur, floaters, and incomplete geometry). Our evaluation is limited to walk-up observations without controlled measures of presence or avatar-related effects. Future work will improve capture density and reconstruction robustness to mitigate artefacts, extend navigation to larger connected environments with comfortable locomotion, and run controlled user studies.

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